

# Air Purity Rectification System: Enhancing Environment

Annlyn Maria Tojy, Joyal J Maliakal, R Devanand, Reswin Geo Thomas, and Anoop Jose Chittilappilly

Jyothi Engineering College, Thrissur

APJ Abdul Kalam Technological University, Kerala, India

Email: joyaljayan06@gmail.com

**Abstract**—The air quality of most under developed and developing countries, as indicated by the Air Quality Index, is generally poor, due to high levels of particulate matter (PM) pollution. Vehicles industries and other human activities contribute to majority of the pollution by emitting various hazardous and toxic chemical pollutants. Even the highly invested filter equipments failed to resolve the issue. This paper is an experimental solution on how to rise to the challenges of recent air filtration mechanisms by combining the adaptive technologies in filtration techniques. Here we introduce a hybrid filtration technique that can eliminate both PM and specific chemical oxide pollutants. A set of composite filters, a sucking mechanism and a concentration technique is used to sort out the pollutants. The electrostatic precipitation used can provide both the desired concentration and time capability for the desired reactions. Whenever the sensors detects the pollutant presence above a certain threshold, the controller activates the circuitry of sucking mechanism and concentration mechanism. The sucked particles are electrostatically charged to attain the concentration on desired filter surface. The filter oxidizes the pollutants and adheres them on the surface.

**Index Terms**—air-purity, rectification, electrostatic precipitator, combination filter, mobile air purifier

## I. INTRODUCTION

We are facing a serious air pollution crisis especially in developing countries, and existing filters just aren't cutting it. The struggle to handle the complex mix of pollutants we find in real-world environments is extreme. As per Air Quality

Index(AQI) most countries possess a poor air quality status even though they initiate high cost filtering mechanisms. Even the highly promising chemical-oxidation technique often falls short. Either the pollutants are too sparse, or the reaction doesn't have enough time to be completed. There should be a way to concentrate the whole pollutants for a desired time onto the filter surfaces for enhanced efficiency.

Mechanically, increasing the concentration and slowing down the flow are the resolving methods that can be initiated, but that introduces its own set of problems. What we need is a clever electronic system that can concentrate the pollutants and give the oxidation process the time it needs to work effectively. Hence the electronic system can cover up the mechanical discrepancies and handle the whole system with maximum stability and efficiency. The survey conducted on existing technologies reveals the drawbacks and highlights the areas for innovations.

## II. LITERATURE REVIEW

Significant challenges found for existing filters include catalyst deactivation, temperature limitation, material degradation and so on. Biological filters carry the potential for microbial growth, which have a high environmental dependency and risk of contamination of existing environment. The challenges point towards the need for new filtration techniques that could possibly eliminate these challenges. The survey was conducted on various existing systems to analyze various sensing mechanisms such as thermal, optical, catalytic, electrochemical, infrared, acoustic and semiconductor sensors and various filtering techniques to identify the most reliable and economical solutions for key challenges [1] [2]. The primary filtration of dust particles is made possible through fine meshes. Filtration mechanisms include indoor air filtering mechanisms of VOC (volatile organic compounds) through activated carbon[5], UVGI (ultraviolet germicidal irradiation) filters, Combination filters using HEPA-UV (high efficiency particulate air - ultra violet) and impregnated carbon meshes[3][4], and various other filtering techniques to filter out each pollutant separately, such as electro-swing reactive adsorption for carbon dioxide in which the reverse polarity applied to carbon nanotubes adheres the carbon dioxide particles[6], catalytic bag filters which makes the flue gases pass through narrow holes created through SCR (selective reduction catalyst) for nitrogen based oxides[7], biological composite filters in which algae is coated on dried sponge[8]. Research was done on how to make the hybrid filter by integrating the most suitable qualities of existing technologies. The drawbacks can be easily solved through oxidizing filters that process through chemical reactions, but the concerns are to specifically attract the pollutants towards the filter and to hold the pollutants until the desired reaction time. When this is made possible, we can ensure the maximum efficiency by ensuring the maximum possibility of reaction. Temperature or pressure limitations can be avoided when the particles are adhered using electrostatic concentration. Atomizers were used to sprinkle off the accumulated dust particles on the prime meshes.

## III. SYSTEM ARCHITECTURE

The figure shows the basic system architecture which integrates the sensors, filtering unit, microcontroller and the feedback unit.

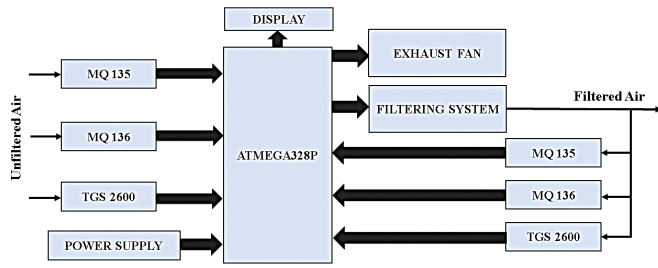


Fig.1. Architecture

#### A. Sensing Mechanisms

The chemical sensors MQ-135, MQ-136, TGS-2600 are used as the sensing terminology (The term MQ denotes metal oxide semi-conductor and TGS denotes tri-glycine sulfate)[9][10][11]. The detection mechanism is the change in conductance while the gases react with the sensor surface. The pollutants react with the semiconductor material inside the sensor to produce a change in conductivity. These sensors provide efficient data on specific pollutants such as metal-oxides of carbon, nitrogen and sulfur. Specifically, MQ-135 detects CO, CO<sub>2</sub>, NO<sub>x</sub>[9]. MQ-136 is the sulfide detector[10]. TGS is a sensor used in OR logic with MQ-135 for detecting CO[11].



Fig.2. Sensors

The sensory signals are passed onto a micro-controller chip (ATMEGA-328) to generate a corresponding control signal. The sensor data such as the presence of pollutants, its concentration and the working condition of the system are shown using an LCD display. The same sensor circuitry is placed as a feedback circuitry also.

#### B. Filtration Technique

The control signal activates the sucking mechanism and the electrostatic precipitator together. A fine mesh is placed before the suction point to avoid the entry of large particles. For the desired reaction time, the precipitator charges the ions to concentrate them towards the filter and holds onto it until the desired time. Hence, the mechanical slowdown of the process causes the instability. A negative terminal of a high voltage transformer of an approximate value of 400kV is connected through a set of fine meshes to charge the pollutants negatively.

The composite filter that initiates the reactions is either made of conductor particles, or the filter is enclosed with fine meshes/ aluminium foil in a zigzag manner or an arranged mesh order. The whole conducting part is connected with the

positive terminal. The negatively charged particles are attracted towards the filter and remain on the filter until the reaction is completely done. This reaction time is specifically calculated and is primarily fed to the micro-controller.

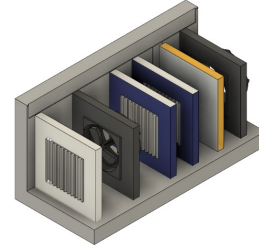


Fig.3. Basic Terminology

The basic terminology shows the skeleton design of the filter architecture. The primary layer is a fine mesh that can filter out dust particles that could damage the inner filters. The second layer shows the primary ground of the concentrator circuit in which the particles are getting negatively charged. The third filter layer is the composite material that is enclosed in a conducting material or frame and is also connected to the positive high-voltage supply at the same time.

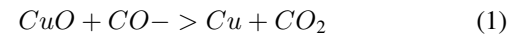
The figure shows two sucking mechanisms which provide sufficient flow and effective air pressure on filter plates. The one in the suction end is an exhaust fan of low CFM (approx 300 cubic feet per minute) which is added to focus the pollutants towards the filter. The one provided at the exhaust side is a centrifugal pump with a high CFM value (approx 1600 CFM)

### IV. COMPOSITE FILTER

A composite material filter consists of two or more elements or compounds mixed in a uniform manner in a specific ratio to perform the reactions to remove the pollutants. In this paper, two composite filters are introduced in which one converts the unstable carbon monoxide into stable carbon dioxide and the other one oxidizes the carbon, nitrogen and sulfur oxides. The filters section includes:

#### A. C-CuO filter

In this filter, activated carbon is impregnated with cupric oxide. This is the primary reactive step of the filtration process. This sets pollutants into a reactive form that can formulate the solidification process. Carbon monoxide, the comparatively instable compound, is oxidized into carbon dioxide, its stable and less toxic form.



Here CuO acts as the oxidizing agent where as carbon provides the surface area for the reactions to occur at a large rate. The filter does its primary role in which it converts partially combusted and highly reactive oxides into its stable form which is comparatively less toxic.

### B. Zephyr-X filter

Zephyr represents clean air where X represents the model number of type of composite filter. Zephyr-X is a specially designed filter by the mixing of potassium permanganate and calcium oxide on a ceramic fabric, under the presence of liquid sodium silicate, a bonding reagent. The material is a powdered or granular solid that can be mixed or impregnated with activated carbon to achieve the desired reaction characteristics. The material mix is melted and coated on the ceramic fabric to achieve the composite section. Carbon provides the necessary surface area for pollutants to adhere.

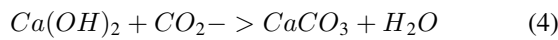
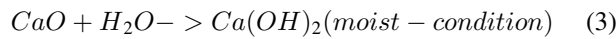
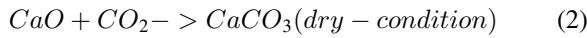
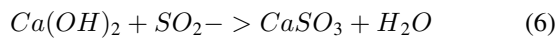
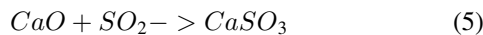


Fig.4. Composite filter

The re-release of the filtered waste into air makes the system inefficient, hence the carbon dioxide can be converted into a powdered carbonate salt by reacting it with the calcium oxide. The collected carbon waste will adhere on the filter surface in a powdered form and can be extracted by sequential processes of chemical reactions. Fabric can potentially support and participate in redox reactions depending on the conditions. The sulfur oxides reacts with calcium oxide to form calcium sulfite, since potassium permanganate ( $KMnO_4$ ) is a strong oxidizing agent it will also oxidize the sulfur oxides into potassium sulfate. Similarly Nitrogen get oxidized too.



But for high reaction rates of the oxidization of sulfur and nitrogen a moist condition is required. Excess moisture leads to clumping of the composite and decrease the overall efficiency.

### V. CIRCUIT DESIGN

The figure shows the detailed electronic circuit diagram that connects the sensor circuit, the filtering particulars, and the display feedback. It shows how the input/output modules are interconnected onto the ATMEGA-328 micro-controller chip. MQ-135 detects the basic carbon compounds whereas TGS-2600 is specific for carbon monoxide.

TGS-2600 added for an additional reliability of sensor panel. MQ-135 is a sulfide gas sensor. More reliable and costly sensors can be integrated for higher performance efficiency.

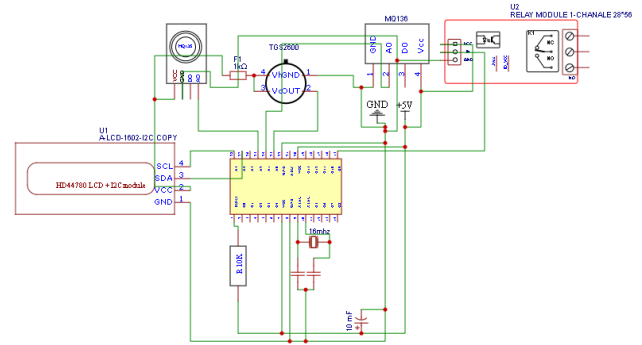


Fig.5. Circuit design using EasyEDA

Two relays are added for the smooth operations of pump and transformer. High voltage transformer supplies a 400 kV for the charging of particles. Pump is used for the inlet air supply. hh Fig.6. PCB design using ALTIUM

This figure is the simplified PCB circuit designed to reduce the complexity and make the design small, precise and efficient.

### VI. RESULT

Electrostatic precipitator concentrates the pollutants towards the filter. Feedback data collection using the sensor circuit at the exhaust end shows the concentration of smoke onto the filter until the circuit is ON. The graphs shows the variations in serial monitor while exposed to gases and the stability time of the used sensors.

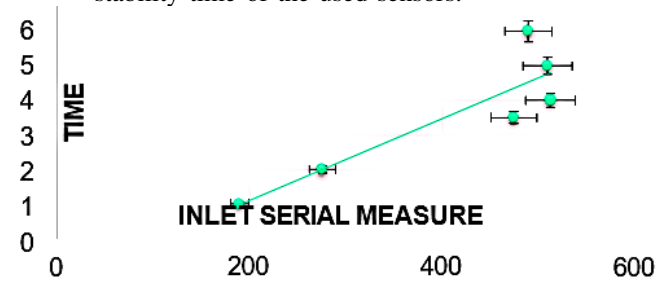


Fig.7. Serial value inlet characteristics

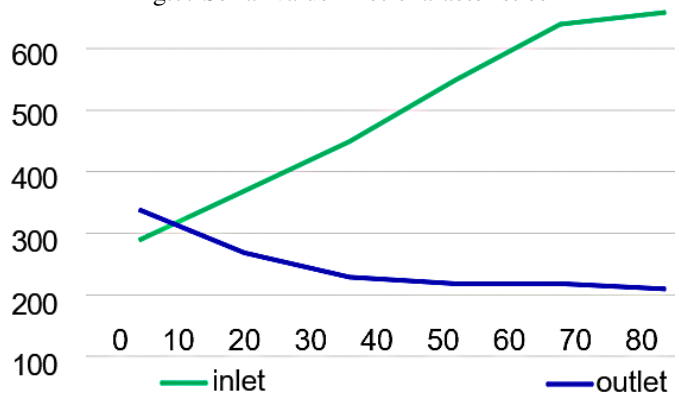


Fig.8. Serial inlet-outlet characteristics

We can set the required reaction time manually through controller programming. The developed composite filter demonstrates the effective pollutant removal through a sequential process of filtration. The result of purification was done on the basis of sensor feedback. The following graphs shows the inlet and outlet values of gases.

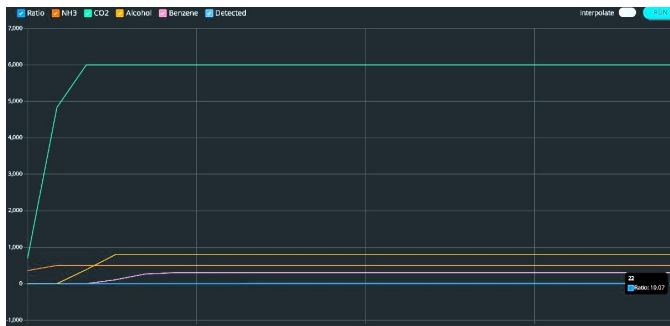


Fig.9. Inlet ppm characteristics

The graph shows the serial PPM values of each gases sensed by the sensor modules in an auditorium, the detection of other gases are due to exposure to perfume.

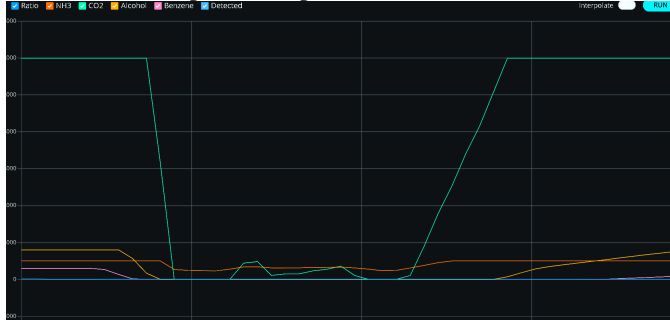


Fig.10. Outlet ppm characteristics

The dipped surface of the outlet graph shows the area where the circuit is on, which facilitates the concentration of pollutants towards the filter. From the graph, we can see the significant reduction of pollutant rate.

## VII. CONCLUSION

The results demonstrate the effectiveness of the capture and retention of pollutants in the filtering process. The integration of composite material technologies and electrostatic concentration allows the optimal oxidation and removal of pollutants. Experimental results validate the statement showing a remarkable reduction of toxicity levels. The use of composite filters ensures the breakdown of harmful gases further enhancing the air purity through oxidation. Additionally, the sensor feedback panel ensures the real-time monitoring making the system reliable. In general, the study demonstrates a viable solution for improved air purity which contributes to a healthier and sustainable future.

## REFERENCES

- [1] S. Panda, S. Mehlatat, N. Dhariwal, A. Kumar, and A. Sanger, "Comprehensive review on gas sensors: Unveiling recent developments and addressing challenges," *Materials Science and Engineering: B*, vol. 308, p. 117616, 2024.

- [2] S. Dhall, B. Mehta, A. Tyagi, and K. Sood, "A review on environmental gas sensors: Materials and technologies," *Sensors International*, vol. 2, p. 100116, 2021.
- [3] A. Dhawale, A. Kamble, and S. Ghule, "Solar powered indoor air purifier with air quality monitor," *IEEE 9th International Conference on Electrical Energy Systems (ICEES)*, 2023.
- [4] R. P. Eldho, R. C.P. D. Rani, A. P. Tiwari, M. Ashraf, and Akanksha, "Design and development of activated carbon-based solar air purifier," *IEEE 4th International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCCMLA)*, 2022.
- [5] D. V. C.H., A. Srividya, S. Madhavi, B. Jayabharathi, and N. Rajeswari, "Outdoor automatic air purification redefined with air quality monitor," *IEEE Proceedings of the 7th International Conference on Inventive Computation Technologies (ICICT)*, 2024.
- [6] S. Voskian and T. A. Hatton, "Faradaic electro-swing reactive adsorption for co2 capture," *Energy Environmental Science*, vol. 12, pp. 3530–3547, 2019.
- [7] C. Li, L. Huangfu, J. Li, and S. Gao, "Recent advances in catalytic filters for integrated removal of dust and NO<sub>x</sub> from flue gas: fundamentals and applications," *Resources chemicals and materials*, vol. 1, pp. 275–289, 2022.
- [8] P. Yewalea, N. Waglea, S. Lenka, et al., "Studies on BIOSMOTRAP: A multipurpose biological air purifier to minimize indoor and outdoor air pollution," *Journal of cleaner production*, vol. 357, pp. 1–17, 2022.
- [9] "Technical data MQ-135 gas sensor," Hanwei Electronics Co., Ltd.
- [10] "MQ-136 semiconductor sensor for sulfur dioxide," Hanwei Electronics Group Corporation.
- [11] TGS-2600 - for the detection of air contaminants," Figaro